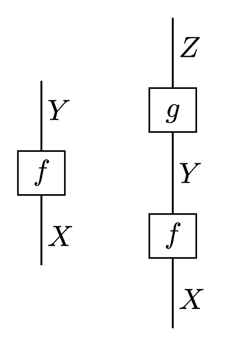
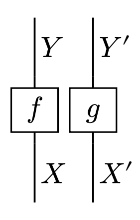
copycat gallery

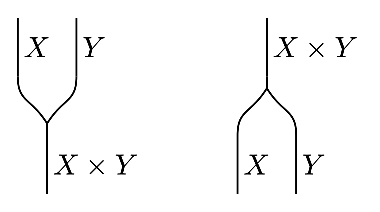
A kernel $f : X \rightsquigarrow Y$, and sequential composition:



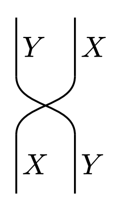
Parallel composition:



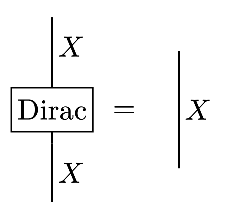
A product wire $X \times Y$ may be drawn as two wires, and back:



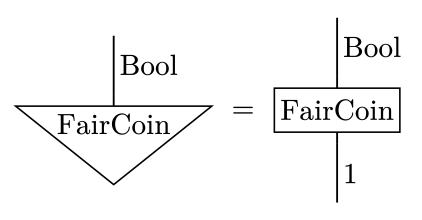
The swap:



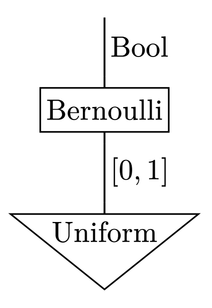
The Dirac kernel is the bare wire:



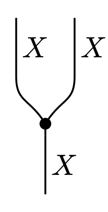
A state is a kernel with trivial input:



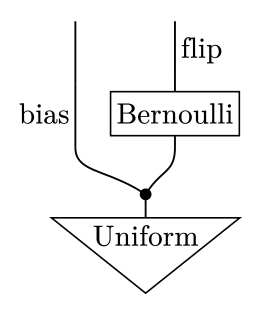
A little probabilistic program:



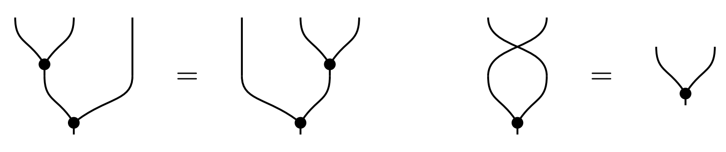
Copying:



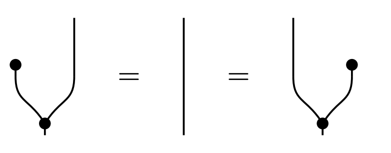
Sampling a bias, then flipping a coin with it:



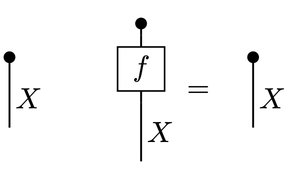
Coassociativity and cocommutativity of copying:



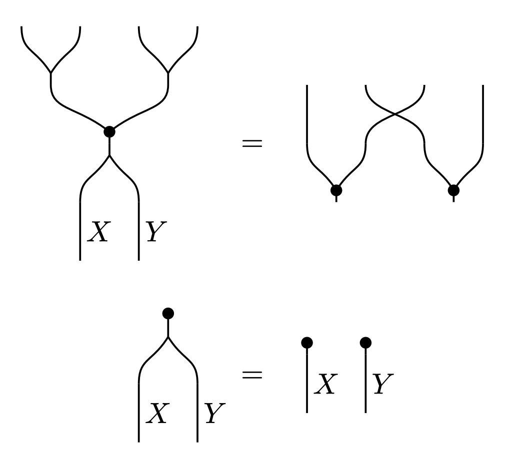
Counitality:



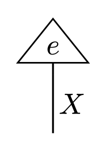
Discarding, and the fact that every kernel is discarded to nothing:



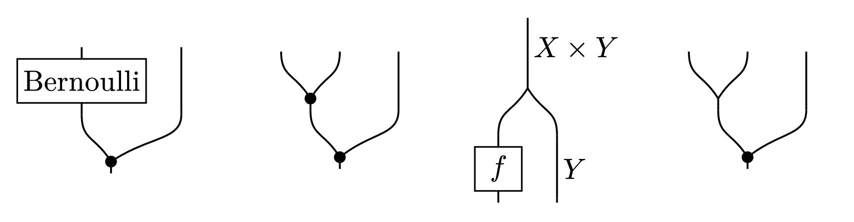
Copying and discarding a product:



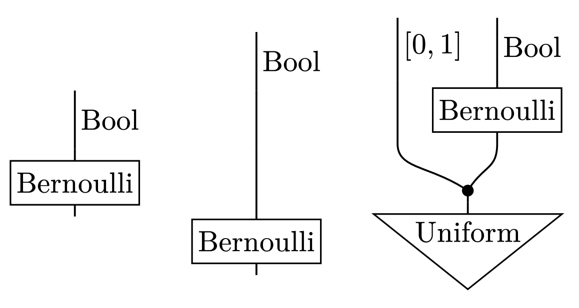
An effect (the mirror image of a state):



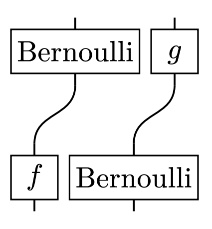
Flexible wiring: the arms of `copy`, `unbundle` and `bundle` run straight to wherever the layer above or below needs them, instead of being bent once inside the fork and once again in a connector band.



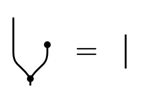
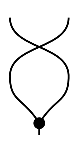
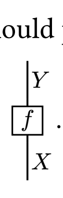
Wires find their own x: a narrow wire layer over a wide box, a stack of three such layers, and the coin-bias program written with the bias wire and the sampled value on separate layers. None of these needs a connector band.



A wire held at different x at both ends bends by itself, with no band inserted:

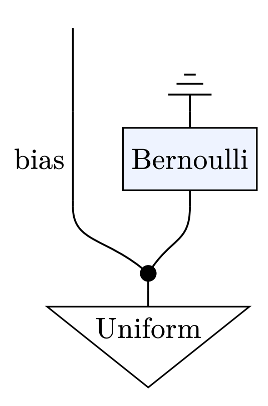


Inline diagrams

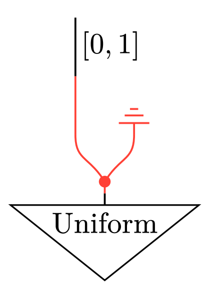
The copy map and the discard map satisfy , so they make every object a comonoid. At the default unit the same term, , is rather large for running text, which is why inline uses should pass a smaller `unit`, and a smaller label size when the diagram carries labels: .

Style overrides

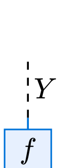
A style override can target a whole diagram (the `style:` argument of `string-diagram`), where element groups inherit the root `stroke` and `fill` and partial strokes fold, cetz-style:



It can target a sub-diagram, via `styled` or the `style:` argument of `serial` and `parallel`:

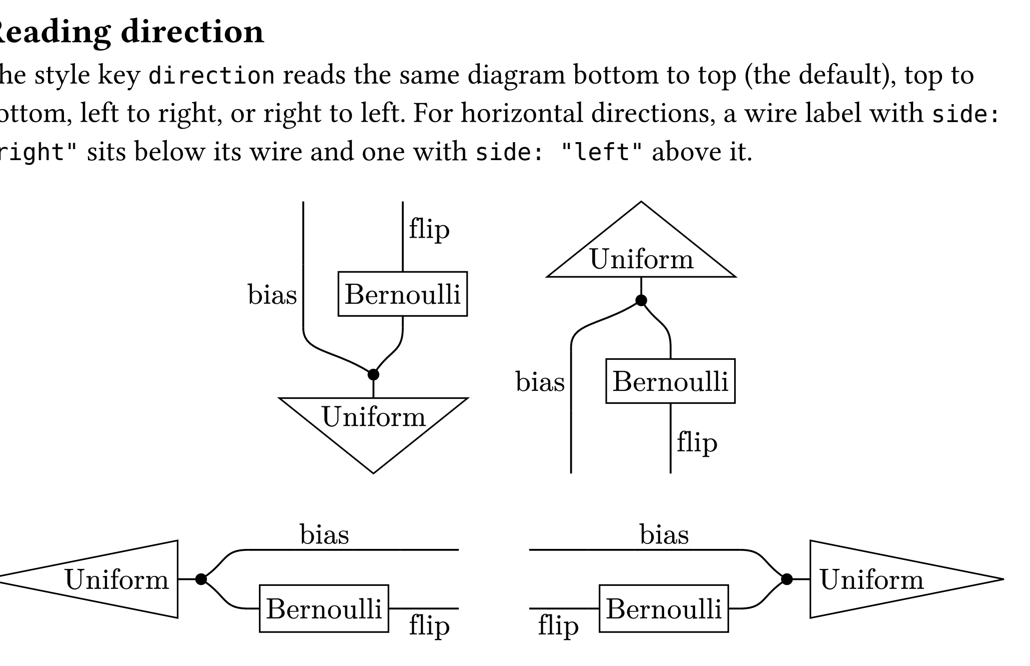


Or it can target a single element, whose inline `stroke` and `fill` restyle just that element:



Reading direction

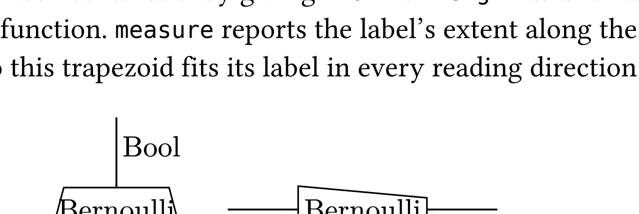
The style key `direction` reads the same diagram bottom to top (the default), top to bottom, left to right, or right to left. For horizontal directions, a wire label with `side: "right"` sits below its wire and one with `side: "left"` above it.



Custom primitives

`primitive` turns a cetz drawing into a rigid element that composes like any other. The drawing function receives the resolved style and the element's geometry, and draws in unit coordinates with the origin at the bottom-left corner.

The snake equation of a compact closed category, and a spider:



An element can size itself to its label by giving `width` or `height` as a function of the style and a `measure` function. `measure` reports the label's extent along the element's width and height, so this trapezoid fits its label in every reading direction: